

GANESH GUPTA

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Experience

Oracle

Jun 2025 – Present

Software Engineer, AI Platform

- Own several agentic engineering-automation workflows (LangGraph, OpenAI Agents SDK) that eliminate manual toil across code review, testing, and release processes.
- Write infrastructure/systems-level code in Go, Rust, and C++ for GPU cluster scheduling, resource allocation, and fault-tolerant placement logic under hardware-scarcity constraints across AWS, GCP, and CoreWeave.
- Built observability and autoscaling for production inference serving via Prometheus/Grafana, sustaining fleet utilization through node loss and provider outages.
- Set engineering standards for CI/CD, containerized deployment, and secure development practices across the platform, working in Agile sprints with Git-based version control.

QureAI (AI Startup)

Mar 2026 – Present

ML Infrastructure Engineer

- Compiled and optimized production LLM inference engines using TensorRT-LLM's build pipeline (LLVM-based compilation tooling), tuning kernel selection and precision for multi-GPU deployment.
- Improved semantic retrieval recall 61% to 89% on a 2K-query eval set by rebuilding the chunking pipeline with hybrid dense-sparse retrieval and re-ranking.

Nomura Research Institute Financial Technologies

Sep 2019 – Aug 2023

Software Engineer

- Architected and operated 25+ production C++ and SQL-backed microservices for 1M+ users, sustaining 99.9% uptime over 4 years on AWS (EC2, RDS, SQS).
- Drove a redesign of nightly data aggregation, cutting processing from 8 minutes to 90 seconds via indexed views and batch partitioning.
- Standardized failure-handling contracts (retry, timeout, propagation semantics) across 25+ services, reducing production escalations 60% in one quarter.
- Mentored engineers through code reviews and design write-ups, driving consistent engineering standards.

Projects

Distributed KV Store & VPC Control Plane Simulator | github.com/ganeshhgupta/distributed-kv-store

- Built a distributed key-value store in C++/Go implementing Raft consensus, linearizable reads, and log compaction; simulated VPC control-plane semantics for network partition and leader-election testing.

RootCause AI | github.com/ganeshhgupta/rootcause-ai

- LangGraph multi-agent system diagnosing production incidents via tool-use chains; routes alerts to specialized sub-agents and synthesizes evidence-backed root cause reports, reducing mean time to resolution.

Watch-your-LLM | github.com/ganeshhgupta/watch-your-llm

- Production LLM eval platform with hallucination detection, drift tracking, rubric-based scoring, and regression alerting across model versions; FastAPI backend, PostgreSQL.

Education

University of Texas at Arlington

Aug 2023 – May 2025

MS, Computer Science, GPA 4.0/4.0

MCKV Institute of Engineering

Jul 2016 – Jul 2020

BTech, Computer Science

Skills

Languages: C++, Go, Rust, Python, TypeScript, SQL, Java

Systems & Compilation: TensorRT-LLM (LLVM-based compilation), Distributed Systems, Raft Consensus, Fault-Tolerant Scheduling

Infra & Observability: GPU Clusters (CUDA, InfiniBand), Slurm, Prometheus, Grafana, Docker, Kubernetes

Cloud: AWS (EC2, S3, RDS, Lambda, SQS), GCP, CoreWeave, CI/CD, GitHub Actions

Certifications

AWS Certified Developer Associate | GCP Professional MLOps Engineer